When Training-Only Risk Calibration Does Not Improve Element-Shift Prediction: A Frozen Negative Study

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Abstract

Composition-only materials models can interpolate accurately while failing under chemical support shift. We tested whether a training-only risk model could identify when a five-seed CatBoost ensemble should be shrunk toward ridge regression. The protocol was frozen before outcome-bearing execution. Within each of three outer element exclusions (Cu, Ag, and B), eleven pseudo-element exclusions generated harm labels without using outer-test targets. Four gate temperatures were evaluated under a fixed selection rule. A private Kaggle run completed 180 CatBoost GPU fits on dual Tesla T4 accelerators. Source and output archives were hash-verified, all 18 hosted tests passed, and an independent local verifier recomputed 156 metrics from retained predictions. The selected gate ($T = 0.40$ eV) achieved element-macro mean absolute error (MAE) of 0.779667 eV versus 0.779536 eV for the identical ungated ensemble, an improvement of -0.000130 eV. The frozen admission criterion required an improvement greater than 0.020 eV. The gate improved Ag and B by less than 0.001 eV each and worsened Cu by 0.001921 eV. Extra Trees was the strongest control at 0.728675 eV macro MAE. A clearly labeled post hoc row-cluster bootstrap gave a 95% percentile interval of $[-0.001964, 0.001428]$ eV for the selected-gate effect. The scientific disposition is `DEVELOPMENT_NOT_SUPPORTED`; protected Li, Zn, Ga, Se, and Te outcomes were not opened. The result shows that moderate pseudo-shift harm ranking did not translate into a practically useful adaptive gate under this frozen design.

1 Introduction

Composition-only property models are inexpensive screening surrogates, but they omit crystal structure, defects, processing, and measurement context. Matbench made several materials-learning tasks and splits reproducible (Dunn et al., 2020); Magpie descriptors provide a widely used composition-level representation (Ward et al., 2016). Learned composition models such as Roost and CrabNet demonstrate substantial predictive utility without structural input (Goodall and Lee, 2020; Wang et al., 2021). That utility does not by itself establish reliability when a chemical element is absent from training.

Domains-of-applicability studies document the degradation that can accompany support shift (Sutton et al., 2020), while redundancy-controlled evaluation can expose optimistic interpolation estimates (Li et al., 2024). A tempting response is to estimate predictive risk from the training data and intervene when risk is high. The difficult question is whether the estimated risk transfers to the particular shift encountered at deployment.

We evaluate a narrow mechanism. A strong direct ensemble and a regularized linear model are fit to the same target-independent descriptors. Pseudo-element exclusions inside the outer training data teach a separate model to predict when the direct ensemble has larger absolute error than ridge. Positive predicted harm exponentially shrinks the direct prediction toward ridge. The study asks whether this adaptive gate improves leave-one-element-out MAE relative to the identical ungated ensemble.

The experiment was prospectively frozen after an earlier distance-gated version showed negligible development benefit and diagnostic weaknesses. The successor changed the representation, direct model, and gate, so it received a new protocol and source identity. Its protected elements remained sealed behind a conjunctive development rule. This report presents the complete verified negative development result rather than continuing to search until a positive gate appears.

2 Methods

2.1 Dataset, representation, and outer shifts

We use Matbench v0.1 `matbench_expt_gap`, containing 4,604 formula–band-gap rows (Dunn et al., 2020). The accepted compressed artifact has SHA-256 `783e7d1461eb83b00b2f2942da4b95fda5e58a0d1ae26b581c24cf8a82ca75b2`. The 270-dimensional target-independent representa-

tion concatenates 132 fixed fraction and periodic-summary features with Magpie elemental-property statistics and six stoichiometric norms. Missing values are imputed using the outer training set only.

Development consists of three declared leave-one-element-out shifts. For element E, every row containing E forms the test set and is excluded from training. The train/test counts are 4,214/390 for Cu, 4,408/196 for Ag, and 4,366/238 for B. The three test sets have 824 memberships over 821 unique Matbench row identifiers because a composition can contain more than one held-out element. Li, Zn, Ga, Se, and Te are protected and were not evaluated.

2.2 Direct model and controls

The primary direct model is a five-seed CatBoost ensemble (Prokhorenkova et al., 2018). Each member uses 1,200 iterations, depth 8, learning rate 0.035, ℓ_2 leaf regularization 5, random strength 0.5, border count 128, and Bayesian bootstrap temperature 1.0. Predictions are clipped only at zero and then averaged. The fixed seeds are 20261021 through 20261025.

Controls are the training median; standardized ridge with internally selected regularization; direct squared-error histogram gradient boosting; a three-seed, 700-tree Extra Trees ensemble (Geurts et al., 2006); one CatBoost seed; the five-seed CatBoost ensemble; and constant convex blends with direct weights 0.25, 0.50, and 0.75. Scikit-learn supplies ridge, histogram boosting, and Extra Trees (Pedregosa et al., 2011). These controls separate adaptive gating from ordinary shrinkage and from model-family choice.

2.3 Training-only pseudo-exclusion calibration

For each outer element, the gate is trained entirely within its outer training set. Eleven declared pseudo elements—Na, Mg, Al, Si, P, Ca, Fe, K, Ti, Mn, and Ni—are excluded one at a time. Ridge and the full five-seed CatBoost ensemble predict each pseudo holdout. For row i, the harm target is

$$h_i = |y_i - d_i| - |y_i - r_i|, \tag{1}$$

where d_i is the direct-ensemble prediction and r_i is the ridge prediction. Positive harm means ridge had smaller absolute error.

A histogram gradient-boosting regressor predicts harm from nine diagnostics. They are the direct mean, ensemble standard deviation, ridge prediction, direct–ridge disagreement, nearest-training distance, unseen-element

fraction, unseen-element count, composition element count, and fraction entropy. Each pseudo element has equal total weight. The pseudo-calibration sets contain 2,069 rows for the Cu outer model, 2,123 for Ag, and 2,105 for B.

Given predicted harm $\widehat{h}(x)$ and temperature T, the direct-model weight and gated prediction are

$$w_T(x) = \exp\left[-\frac{\max\{\widehat{h}(x), 0\}}{T}\right], \tag{2}$$

$$\widehat{y}_T(x) = \max\{0,\ r(x) + w_T(x)[d(x) - r(x)]\}. \tag{3}$$

Thus nonpositive predicted harm preserves CatBoost, while positive predicted harm shrinks continuously toward ridge. The complete temperature grid is $T \in \{0.05, 0.10, 0.20, 0.40\}$ eV.

2.4 Frozen selection and stopping rule

The primary metric is the unweighted mean of Cu, Ag, and B MAE. The numerical minimum is identified first; candidates within 0.005 eV choose the largest T, the least aggressive shrinkage. Protected confirmation is admitted only if all of the following hold:

1. selected-gate macro MAE improves over the identical ungated CatBoost ensemble by more than 0.020 eV;
2. at least two of three development elements improve;
3. the gate is active on 1–90% of every element holdout;
4. CatBoost predictions are nonconstant, every pseudo exclusion completes, and all retained evidence is finite and manifest-valid.

The rule is conjunctive. Failure terminates v2 without opening protected outcomes. The primary effect is

$$\Delta = \frac{1}{3}\sum_{E}\left[\mathrm{MAE}_E(d) - \mathrm{MAE}_E(\widehat{y}_T)\right], \quad E \in \{\mathrm{Cu}, \mathrm{Ag}, \mathrm{B}\}. \tag{4}$$

Positive values favor the gate.

2.5 Evidence and computation

The frozen source commit is `5a56bb5dc97619730f0fbc6544490cdc69f2c109`; the protocol SHA-256 is `c4aa306c7259238075db18ced236e99d59101d57321e733e84c458f2b7237c2e`. The private Kaggle kernel ran Python 3.13.15, CatBoost 1.2.10, matminer 0.10.1, and NumPy 2.5.4 on two Tesla T4 GPUs with 15,360 MiB each. The design requires 60 CatBoost fits per outer element: 11 pseudo exclusions times five seeds plus five outer fits, for 180 fits total.

The run retained member predictions, pseudo-calibration rows, gate weights, per-example predictions, aggregate metrics, feature schema, environment, resource observations, GPU identities, and a content manifest. The published archive checksum was verified before safe extraction. A separate local verifier then checked all files and recomputed every reported metric from the arrays.

2.6 Post hoc descriptive uncertainty

The frozen gate decision uses the practical-effect rule above. After that decision, we computed a descriptive 20,000-resample percentile bootstrap with seed 20261021. The sampling unit is the unique Matbench row identifier across all three holdouts; resampling a row therefore keeps its memberships coupled. This analysis was not prospectively specified, and the three development elements are selected scenarios rather than a random population of elements. It is reported as a sensitivity description, not as a new decision rule.

3 Results

3.1 Primary development decision

All 180 GPU fits completed in 6,845.90 seconds. Peak host resident memory was 914.55 MiB and the pageout delta was zero. All 18 hosted tests passed. The gate was active on 18.7% of Cu, 13.3% of Ag, and 19.3% of B, satisfying the nondegeneracy rule. CatBoost predictions were nonconstant, and all 33 declared pseudo exclusions completed.

The selected $T = 0.40$ gate achieved macro MAE 0.779667 eV; the ungated CatBoost ensemble achieved 0.779536 eV. Hence $\Delta = -0.000130$ eV. Per-element effects were -0.001921 eV for Cu, $+0.000967$ eV for Ag, and $+0.000563$ eV for B. The sign-count condition passed, but the required

Table 1: Development MAE (eV). Macro is the unweighted mean across elements. The Extra Trees result is a secondary control; the frozen hypothesis concerns the selected risk gate versus the identical CatBoost ensemble.

Model	Cu	Ag	B	Macro
Extra Trees ensemble	0.7271	0.3708	1.0881	0.7287
CatBoost single seed	0.6978	0.4019	1.2164	0.7720
Direct HGB	0.6460	0.4395	1.2329	0.7728
Risk gate, $T = 0.05$	0.7078	0.3950	1.2266	0.7765
CatBoost ensemble	0.7095	0.3991	1.2300	0.7795
Selected risk gate, $T = 0.40$	0.7115	0.3981	1.2294	0.7797
Ridge	0.8021	0.5484	1.3216	0.8907

> 0.020 eV practical effect failed. The prospectively frozen disposition is therefore `DEVELOPMENT_NOT_SUPPORTED`.

The numerical minimum was $T = 0.05$ at 0.776451 eV, a 0.003086 eV improvement over CatBoost. It was within 0.005 eV of the $T = 0.40$ candidate, so the tie rule selected $T = 0.40$. Neither candidate approached the admission threshold.

3.2 Controls and risk transfer

Extra Trees was the best evaluated control at 0.728675 eV macro MAE, improving by 0.050862 eV relative to the CatBoost ensemble. One CatBoost seed (0.772015 eV) and direct histogram boosting (0.772843 eV) also outperformed the five-seed CatBoost mean. These are development comparisons under the same three shifts; they do not establish a leaderboard claim or authorize protected evaluation.

The harm model's Spearman correlation on pseudo-calibration rows was 0.666 for Cu, 0.656 for Ag, and 0.707 for B. On the actual outer holdouts, descriptive risk–harm correlations fell to 0.309, 0.527, and 0.240. The gate therefore learned nontrivial ordering within pseudo shifts, but that ordering transferred weakly and inconsistently to the declared outer shifts.

3.3 Post hoc uncertainty description

The row-cluster bootstrap interval for the selected-gate macro improvement was $[-0.001964, 0.001428]$ eV. The fraction of bootstrap replicates above

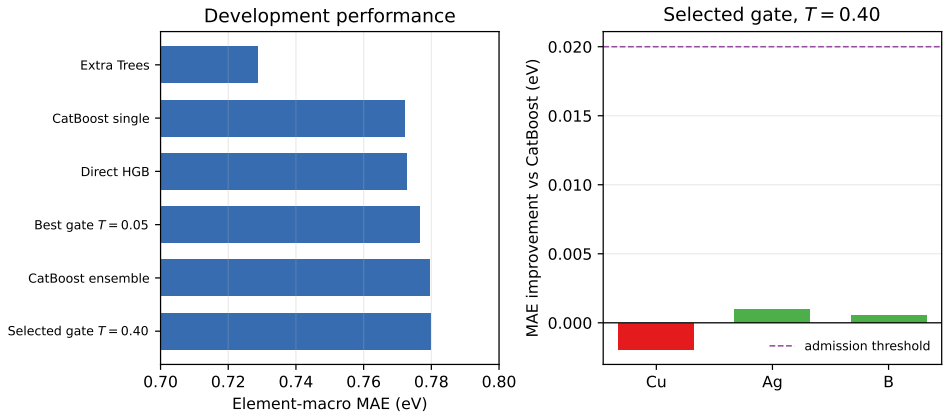


Figure 1: Left: development macro MAE for principal models (axis begins at 0.70 eV). Right: selected-gate paired MAE improvements by element; positive values favor the gate. The dashed line is the 0.020 eV macro admission threshold, shown as a scale reference.

Table 2: Risk-calibration diagnostics. Outer correlations are descriptive because outer targets participate in evaluation.

Outer element	Pseudo rows	Pseudo Spearman	Outer Spearman
Cu	2,069	0.666	0.309
Ag	2,123	0.656	0.527
B	2,105	0.707	0.240

zero was 0.4596; none of 20,000 exceeded 0.020 eV. Paired-row intervals were $[-0.006702, 0.001790]$ eV for Cu, $[-0.001664, 0.003509]$ eV for Ag, and $[-0.000391, 0.001447]$ eV for B. These descriptive intervals agree with the primary conclusion that any effect under this development design is tiny relative to the declared practical threshold.

3.4 Independent verification

Kaggle reported the kernel terminal state as complete. The source roundtrip matched 40 files with no missing, extra, or changed entries. The output archive SHA-256 was `cf8aecebf1dec60ccfb48445b25a891b75b2feff7fb37cf0e08774fbdeeb420a`. Its 18 members had no absolute paths, traversal components, links, or device entries. All 17 extracted

run files matched the direct Kaggle download byte-for-byte; their retained tree digest was `43eb835f5fe7eef682502fee15c3e6fd3326587313fd2a93a6a5fe398cf32c5c`. The independent verifier found a valid manifest and reproduced 156 metric comparisons. Both hosted and local verification confirmed that protected and official-fold outcomes were absent.

4 Discussion

The result separates risk ranking from useful intervention. Pseudo-exclusion correlations of roughly 0.66–0.71 show that the harm model captured structure in its constructed calibration set. The much smaller outer correlations and near-zero MAE effects show that this structure did not support a practically valuable gate on Cu, Ag, and B. A risk score can be statistically ordered yet still fail to improve the downstream decision.

The constant-blend controls also matter. They performed worse than CatBoost, which means indiscriminate shrinkage toward ridge was harmful. The adaptive gate avoided most of that damage, but avoidance is not improvement. Meanwhile Extra Trees substantially outperformed the CatBoost ensemble on these development shifts, suggesting that model-family robustness was more consequential than the proposed risk correction in this experiment.

The pseudo-exclusion data include extreme ridge failures, especially for Fe. Because pseudo elements received equal total weight, those retained cases may have shaped the learned harm surface. Removing, clipping, or reweighting them after observing the result would be a new hypothesis. Such work must use a new versioned protocol and cannot reinterpret v2 as successful.

5 Limitations and Claim Boundary

Only three development elements were evaluated, and their test sets overlap. They are fixed stress scenarios, not a random sample of chemical elements, so neither the bootstrap nor the sign count supports population-wide inference. Composition-only descriptors cannot resolve polymorphs, defects, processing, or measurement context. The public dataset itself may contain redundancy, and leave-one-element-out shift does not represent every deployment shift.

GPU boosting can vary with floating-point reduction order. We retained all five member predictions and report the executed ensemble rather than

claiming byte-identical retraining. Hyperparameters were fixed and no protected outcome was used, but selecting among four temperatures on three outer scenarios can still overfit development. The admission threshold was intended to prevent a small selected fluctuation from reaching confirmation.

The result supports one bounded conclusion: the frozen v2 risk gate did not justify protected confirmation on Cu/Ag/B. It does not prove that risk gating is universally ineffective, that Extra Trees is state of the art, or that any model identifies stable, synthesizable, safe, novel, or useful materials. No v2 claim is available about official Matbench-fold preservation or the protected Li, Zn, Ga, Se, and Te shifts.

6 Conclusion

A fully executed, hash-verified GPU experiment found no practically meaningful benefit from training-only risk-gated shrinkage of a CatBoost ensemble. The selected gate slightly worsened macro MAE, and even the numerically best gate fell far below the prospectively frozen improvement threshold. The correct scientific action is to retain the negative result and stop before protected confirmation. This outcome narrows future work toward better shift proxies, robust treatment of pathological pseudo exclusions, and model-family choices, all under separately versioned protocols.

Reproducibility and Artifact Statement

The retained package contains the frozen protocol; exact source commit and archive digests; raw outer predictions; five CatBoost members per element; pseudo-calibration arrays; per-example risk features and gate weights; GPU admission and postflight receipts; environment freeze; resource trace; kernel log; aggregate results; and a manifest. The private Kaggle output was downloaded from kernel version 1 and independently verified locally. The deterministic secondary analysis regenerates Table 1, Figure 1, and the post hoc uncertainty receipt. At the time of this report, the hosted kernel and dataset are private; the retained local artifacts are the review record and are not described as independently public.

Data and Ethics Statement

The study uses the previously published Matbench experimental-gap benchmark and contains no human participants or personal data. It makes no

recommendation to synthesize or deploy a material. Experimental validation, safety assessment, and application-specific review remain necessary for any downstream use.

References

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A Frozen Identity and Decision Matrix

Item	Value
Study	`sparking-development-v2-risk-gated-gpu`
Run	`kaggle-development-v2-r1`
Source commit	`5a56bb5dc97619730f0fbc6544490cdc69f2c109`
Protocol digest	`c4aa306c7259238075db18ced236e99d59101d57321e733e84c458f2b7237c2e`
Selected model	risk gate, $T = 0.40$ eV
Primary effect	-0.000130 eV
Required effect	> 0.020 eV
Disposition	`DEVELOPMENT_NOT_SUPPORTED`
Protected outcomes	unopened

B Exact Verification Commands

```
.venv/bin/python scripts/verify_successor_v2.py \
  runs/kaggle-development-v2-r1
MPLCONFIGDIR=/tmp/sparking-matplotlib \
  .venv/bin/python scripts/analyze_successor_v2.py
```